

How To Design IoT SYSTEMS

PAROMIK CHAKRABORTY

A quick guide to designing a perfect Internet of Things (IoT) system taking into account performance, connectivity, power consumption and security issues

The Internet of Things (IoT) is no longer a technology of the future. Smart cities, connected industries and smart households have indeed ushered in an era where machines can communicate. The beauty of this technology lies in the fact that the complex backend structure of systems is represented to the end-user in the simplest possible form. This requires profound design know-how.

What the IoT is made up of

The IoT can be designed at different scales for different uses. It can start from our homes with simple lighting or appliance control, and expand into the realm of factories and industries with automated machines, smart security systems and central management systems—called connected factories. It has scaled up to entire cities with smart parking, smart metering, waste management, fire control, traffic management and any similar functions involved. However, irrespective of the scale of application, the main IoT backbone remains similar.

The IoT architecture is multi-layered with delicate components intricately connected to each other. It starts with sensors, which are the source of data being collected. Sensors pass data onto an adjacent edge device, which converts data into readable digital values and stores

these temporarily. When the edge senses a suitable wireless network or the Internet, it pushes the locally stored data to a cloud server involved in the application. The data is processed, analysed, stored and forwarded to the end-user device, represented by an application software. All the design fundamentals and challenges revolve around these layers.

Designing the perfect system

Successfully setting up a complete IoT chain comes with certain challenges. Designers need to walk the extra mile to meet all business requirements like prolonged battery life and low power consumption, secured network gateways, ruggedised hardware and more. Electromagnetic compliance (EMC) and resistance to electromagnetic interference (EMI) are also essential. Introducing new devices within a wireless radius without disturbing the existing system is one of the most demanding skills for an IoT design.

Components like the microcontroller unit (MCU), sensors and wireless network are built and handled with special care for an optimal system design. Based on the scale of application, the design is bound to alter. While basics remain the same, minute detailing into engineering improves a system build by leaps and bounds.

Optimising the MCU. An MCU is the central nervous system of the IoT setup. Data collected by sensor nodes is processed by the MCU using software programs, enabling the system to respond accordingly. The MCU for an IoT system is chosen based on application-specific requirements.

For basic applications with limited amount of data to be transmitted every day, an 8-bit MCU will suffice. On the other hand, large-scale connected systems require 32-bit controllers. As 32-bits MCUs have a bigger RAM size and larger flash that can hold a complete network stack and application codes, these are optimal for embedded systems that will apply radio-frequency (RF) program stacks or complex algorithms.

According to Kavita Char, formerly

